



Qualification Pack

Machine Operator - Plastics Extrusion

QP Code: CPC/CAP/Q0301

Version: 1.0

NSQF Level: 3.5

CENTRAL INSTITUTE OF PETROCHEMICALS ENGINEERING & TECHNOLOGY || CIPET Head Office, T.V.K.
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CPC/CAP/Q0301: Machine Operator - Plastics Extrusion

Brief Job Description

The Machine operator helps in handling plastics granules (raw material), operates the Plastic Extrusion Moulding machines, set process parameter, carryout basic quality check of finished products & stores in the desired place

Personal Attributes

Technical aptitude, Safety-conscious, Reliability and punctuality, Organizational skills, Problem-solving ability, Teamwork and communication, Quality-focused mindset, Willingness to learn

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CPC/CAP/N0301: Understand the Basics of Plastics Raw Material & concept of Extrusion Techniques](#)
2. [CPC/CAP/N0302: Understand the basics & importance of Plastics Compounding /Mixing](#)
3. [CPC/CAP/N0303: Perform HDPE /PVC Pipe Extruder Machine Operation](#)
4. [CPC/CAP/N0304: Perform - Plastic Film Extruder Machine Operation](#)
5. [CPC/CAP/N0306: To - carry out reporting and documentation](#)
6. [CPC/CAP/N0307: To carryout basic quality check of finished products with reference to approved product](#)
7. [CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S](#)
8. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
9. [CPC/CAP/N0305: To Carryout House Keeping](#)

Qualification Pack (QP) Parameters

Sector	Chemical & PetroChemical
Sub-Sector	Petrochemicals
Occupation	Plastics Extrusion

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Country	India
NSQF Level	3.5
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8142.0400
Minimum Educational Qualification & Experience	11th Class with NA of experience OR 10th grade pass with 1.5 years of experience OR 8th Class pass with 4.5 years of experience OR Previous relevant Qualification of NSQF Level with 1.5 years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	25/05/2028
NSQC Approval Date	26/05/2025
Version	1.0
Reference code on NQR	QG-3.5-CP-04119-2025-V2-CIPET
NQR Version	2.0

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CPC/CAP/N0301: Understand the Basics of Plastics Raw Material & concept of Extrusion Techniques

Description

This NOS is about providing a basic knowledge of Plastics Raw Materials Used in Extrusion Process and The Extrusion Process.

Scope

The scope covers the following :

- This unit/task covers the following:
- Understanding Polymers & Plastics
- Common Plastics Materials Used in Extrusion Process
- Types of Extrusion Items Produce
- Extrusion Process
- Types of Extruders

Elements and Performance Criteria

Plastics Raw Material

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding Types of Plastics used in Extrusion and Its Properties
- PC2.** Selection Plastics Raw Materials based on the Items Produced
- PC3.** Storing and Handling of Raw Materials and House Keeping

Extrusion Process

To be competent, the user/individual on the job must be able to:

- PC4.** Types of Extruders Used in the Extrusion Process and their Parts
- PC5.** Types of Dies Used for different Extruded Products
- PC6.** Haul Off Units
- PC7.** Storing and Handling of Finished Products and House Keeping

Compounding and Mixing

To be competent, the user/individual on the job must be able to:

- PC8.** Types of Additives, Master Batches
- PC9.** Types of Mixing and Compounding
- PC10.** Measurement of Additives and Materials and Maintaining Formulations
- PC11.** Storing, Handling of Raw Materials in Compounding Area and House Keeping

Health & Safety

To be competent, the user/individual on the job must be able to:

- PC12.** Understanding Safety Equipments and Its Use
- PC13.** Do's and Don't in Area of Operation

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PC14. Safety Precaution Majors before Operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the Organization Chart, Relevant legislation, standards, policies, procedures & Code of Conduct followed in the company
- KU2.** Different Types of Products Manufactured by the Company
- KU3.** Quality Check and Grading and Segregation of non conforming Products
- KU4.** Risk and impact of not following defined procedures/work instructions
- KU5.** Types of documentation in organization and importance of the same
- KU6.** Records to be maintained and implications of non-maintenance of the same
- KU7.** Importance of housekeeping & good shop floor practices (eg. 3S & 5S)
- KU8.** Health, Safety and Environment guidelines
- KU9.** Impact of poor practices on health, safety and environment
- KU10.** Potential hazards and actions to minimize the same
- KU11.** Impact of various practices on cost, quality, productivity, delivery and safety
- KU12.** Handover/ Takeover the equipment/ work area as per company SOP
- KU13.** The Emergency Stops Procedure for Compounding and Extruder Machine
- KU14.** Cleanliness and safety requirements for commencing Compounding & Extruder
- KU15.** Importance of RPM, Temperature and Pressure Parameters
- KU16.** Preparation of Formulations
- KU17.** Raw Material Feeding Auto / Manual Process
- KU18.** Health hazards of process and compounding ingredients
- KU19.** Measurement techniques using gauges and balance (for thickness, width and weight)
- KU20.** Response to emergencies e.g. Power failures, fire and system failures and manual intervention to avoid disaster
- KU21.** Knowledge of appropriate batch size with respect to appropriate machinery

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS2.** Perform functional mathematical operations, including apply basic mathematical principles, such as numbers and space, and techniques such as estimation and approximation, for practical purposes
- GS3.** How to enter into the history card details of the fault identified in the plastic product manufactured

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- GS4.** Read and understand manuals, Warning, health and safety instructions, Raw Material Labels, Components, memos, reports, job cards etc
- GS5.** Read images, graphs, diagrams
- GS6.** Understand the various coding systems as per company norms
- GS7.** Express Opinions or Information clearly
- GS8.** Discuss task lists, schedules, and work-loads with co-workers
- GS9.** Communicate with Supervisors Job & Issue related tasks
- GS10.** Communication between Upstream and Downstream teams
- GS11.** Avoid using jargon, slang or acronyms when communicating with a supervisor /fellow subordinates etc unless it is required
- GS12.** Work in a team and other behavioral skills required to support the small group activities (Quality Circle, Cross Functional Team, Suggestion Scheme)
- GS13.** Practice honesty with respect to company property and time
- GS14.** Take responsibility for completing one own work assignment
- GS15.** Take initiative to enhance/learn skills in one area of work
- GS16.** The capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse one learning
- GS17.** Is open to new ways of doing things
- GS18.** The capacity to envisage and articulate personal goals to develop strategies and take action to achieve them
- GS19.** Avoid absenteeism
- GS20.** Act objectively , rather than impulsively or emotionally when faced with difficult/stressful or emotional situations
- GS21.** Work in disciplined factory environment
- GS22.** Be punctual
- GS23.** Make decisions pertaining to the concerned area of work
- GS24.** Operating different types of extruders
- GS25.** Handling of Plastics Raw Materials, Additives and Master Batches
- GS26.** Handling of various types of material handling equipment like forklifts, trolley etc
- GS27.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS28.** Use common sense and make judgments during day to day basis
- GS29.** Use reasoning skills to identify and resolve basic problems
- GS30.** Identify immediate or temporary solutions to resolve delays
- GS31.** DiagLOe common problems in the machine based on visual inspection, sound etc
- GS32.** Suggest improvements(if any) in process based on experience

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Plastics Raw Material</i>	6	9	-	-
PC1. Understanding Types of Plastics used in Extrusion and Its Properties	2	3	-	-
PC2. Selection Plastics Raw Materials based on the Items Produced	2	3	-	-
PC3. Storing and Handling of Raw Materials and House Keeping	2	3	-	-
<i>Extrusion Process</i>	7	12	-	-
PC4. Types of Extruders Used in the Extrusion Process and their Parts	2	3	-	-
PC5. Types of Dies Used for different Extruded Products	2	3	-	-
PC6. Haul Off Units	1	3	-	-
PC7. Storing and Handling of Finished Products and House Keeping	2	3	-	-
<i>Compounding and Mixing</i>	4	12	-	-
PC8. Types of Additives, Master Batches	1	3	-	-
PC9. Types of Mixing and Compounding	1	3	-	-
PC10. Measurement of Additives and Materials and Maintaining Formulations	1	3	-	-
PC11. Storing, Handling of Raw Materials in Compounding Area and House Keeping	1	3	-	-
<i>Health & Safety</i>	3	7	-	-
PC12. Understanding Safety Equipments and Its Use	1	3	-	-
PC13. Do's and Don't in Area of Operation	1	2	-	-
PC14. Safety Precaution Majors before Operations	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	20	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0301
NOS Name	Understand the Basics of Plastics Raw Material & concept of Extrusion Techniques
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Extrusion, Plastics Extrusion
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0302: Understand the basics & importance of Plastics Compounding /Mixing

Description

This NOS is about maintain a Safe and Healthy work place, adhering to the safety guidelines in the working area

Scope

The scope covers the following :

- This unit/task covers the following:
- Identifying and reporting of risks
- Creating and sustaining a safe, clean and environment friendly work place
- Action taken in the case of emergency and fatal situations

Elements and Performance Criteria

Understanding Chemicals, Additives and Colorants

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding Types of Chemicals, Additives and Colorants.
- PC2.** Properties and Importance of Chemicals, Additives and Colorants and Pigments.
- PC3.** Properties changes by adding Chemicals, additives and pigments in Plastics
- PC4.** Formulation laid down for different Products

Types of Blenders and Mixers

To be competent, the user/individual on the job must be able to:

- PC5.** Types of Blenders , Mixers and their Parts
- PC6.** Preparation Batches as per the formulations
- PC7.** Loading and unloading of Batches

Parameters Involved in High Speed Mixing and Blending

To be competent, the user/individual on the job must be able to:

- PC8.** Temperature, Pressure and Speed involved in Blenders and Mixers
- PC9.** Importance of each and every Parameters
- PC10.** Precaution to be taken care during the batch preparation.
- PC11.** Storing of batches after preparation

Health & Safety

To be competent, the user/individual on the job must be able to:

- PC12.** Understanding Safety Equipments and Its Use
- PC13.** Do's and Don't in Area of Operation
- PC14.** Safety Precaution Majors before Operations

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different Types of formulation vis-a-vis Products Manufactured by the Company
- KU2.** Quality Check and Grading and Segregation of non conforming Batches
- KU3.** Risk and impact of not following defined procedures/work instructions
- KU4.** Types of documentation in organization and importance of the same
- KU5.** Records to be maintained and implications of non-maintenance of the same
- KU6.** Importance of housekeeping & good shop floor practices (eg. 3S & 5S)
- KU7.** Health, Safety and Environment guidelines
- KU8.** Impact of poor practices on health, safety and environment
- KU9.** Potential hazards and actions to minimize the same
- KU10.** Impact of various practices on cost, quality, productivity, delivery and safety
- KU11.** Handover/ Takeover the equipment/ work area as per company SOP
- KU12.** The Emergency Stops Procedure for Mixing and Blending Machines
- KU13.** Machine Cleanliness and safety requirements for commencing Compounding Mixing & Blender.
- KU14.** Importance of RPM, Temperature and Pressure Parameters.
- KU15.** Importance of Formulations Vis-a-Vis Items produced.
- KU16.** Health hazards of process and compounding ingredients
- KU17.** Measurement techniques using gauges and balance
- KU18.** Response to emergencies e.g. Power failures, fire and system failures and manual intervention to avoid disaster
- KU19.** Knowledge of appropriate batch size with respect to appropriate machinery

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS2.** Perform functional mathematical operations, including apply basic mathematical principles, such as numbers and space, and techniques such as estimation and approximation, for practical purposes
- GS3.** How to enter into the history card details of the fault identified in the plastic product manufactured
- GS4.** Read and understand manuals, Warning, health and safety instructions, Raw Material Labels, Components, memos, reports, job cards etc
- GS5.** Read images, graphs, diagrams
- GS6.** Understand the various coding systems as per company norms
- GS7.** Express Opinions or Information clearly

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- GS8.** Discuss task lists, schedules, and work-loads with co-workers
- GS9.** Communicate with Supervisors Job & Issue related tasks
- GS10.** Communication between Upstream and Downstream teams
- GS11.** Avoid using jargon, slang or acronyms when communicating with a supervisor /fellow subordinates etc unless it is required
- GS12.** Work in a team and other behavioral skills required to support the small group activities (Quality Circle, Cross Functional Team, Suggestion Scheme)
- GS13.** Practice honesty with respect to company property and time
- GS14.** Take responsibility for completing one own work assignment
- GS15.** Take initiative to enhance/learn skills in one area of work
- GS16.** The capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse one learning.
- GS17.** Is open to new ways of doing things
- GS18.** The capacity to envisage and articulate personal goals to develop strategies and take action to achieve them
- GS19.** Avoid absenteeism
- GS20.** Act objectively , rather than impulsively or emotionally when faced with difficult/stressful or emotional situations
- GS21.** Work in disciplined factory environment
- GS22.** Be punctual
- GS23.** Make decisions pertaining to the concerned area of work
- GS24.** Operating different types of Blenders and Mixers
- GS25.** Handling of Plastics Additives, Colorants and Master Batches
- GS26.** Handling of various types of material handling equipment like forklifts, trolley etc
- GS27.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS28.** Use common sense and make judgments during day to day basis
- GS29.** Use reasoning skills to identify and resolve basic problems
- GS30.** Identify immediate or temporary solutions to resolve delays
- GS31.** DiagLOe common problems in the machine based on visual inspection, sound etc
- GS32.** Suggest improvements(if any) in process based on experience

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Chemicals, Additives and Colorants</i>	7	10	-	-
PC1. Understanding Types of Chemicals, Additives and Colorants.	1	2	-	-
PC2. Properties and Importance of Chemicals, Additives and Colorants and Pigments.	1	2	-	-
PC3. Properties changes by adding Chemicals, additives and pigments in Plastics	2	2	-	-
PC4. Formulation laid down for different Products	3	4	-	-
<i>Types of Blenders and Mixers</i>	7	12	-	-
PC5. Types of Blenders , Mixers and their Parts	3	4	-	-
PC6. Preparation Batches as per the formulations	2	4	-	-
PC7. Loading and unloading of Batches	2	4	-	-
<i>Parameters Involved in High Speed Mixing and Blending</i>	10	16	-	-
PC8. Temperature, Pressure and Speed involved in Blenders and Mixers	3	4	-	-
PC9. Importance of each and every Parameters	3	4	-	-
PC10. Precaution to be taken care during the batch preparation.	2	4	-	-
PC11. Storing of batches after preparation	2	4	-	-
<i>Health & Safety</i>	6	12	-	-
PC12. Understanding Safety Equipments and Its Use	2	4	-	-
PC13. Do's and Don't in Area of Operation	2	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Safety Precaution Majors before Operations	2	4	-	-
NOS Total	30	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0302
NOS Name	Understand the basics & importance of Plastics Compounding /Mixing
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Extrusion, Plastics Extrusion
NSQF Level	3.5
Credits	3
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0303: Perform HDPE /PVC Pipe Extruder Machine Operation

Description

This OS unit is about knowledge of HDPE / PVC Pipe Extrusion Process.

Scope

The scope covers the following :

- This unit/task covers the following:
- HDPE / PVC Pipe Extruders and their parts
- Extrusion Process and Parameters.
- Machine Operation and Controls.
- Common faults and remedies.
- Health and Safety

Elements and Performance Criteria

HDPE / PVC Pipe Extruders and their parts

To be competent, the user/individual on the job must be able to:

- PC1.** Types of HDPE / PVC Extruders
- PC2.** Extruder Parts and Their Functions
- PC3.** Pressure and Vacuum Sizing Units
- PC4.** Types of Dies Used for different Extruded Pipes
- PC5.** Operations of Haul Off Units
- PC6.** Dismantling and assembling Extruder Parts
- PC7.** Safety Precaution taken during assembling and disassembling

Extrusion Process and Parameters

To be competent, the user/individual on the job must be able to:

- PC8.** Common Process Parameter like Temperature, Pressure and Speed and its controls
- PC9.** Effect of process parameters on Product Properties

Machine Operation and Controls

To be competent, the user/individual on the job must be able to:

- PC10.** Trial Production and checking product stabilization
- PC11.** Actual Production and Parameter / Process Control
- PC12.** Quality Check and Continuous Production
- PC13.** Post production and storing

Common faults and remedies

To be competent, the user/individual on the job must be able to:

- PC14.** Common faults found and trouble shooting
- PC15.** Segregation of faulty product and action taken

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PC16. Disposal of faulty products as per laid down procedure

Health & Safety

To be competent, the user/individual on the job must be able to:

PC17. Understanding Safety Equipments and Its Use

PC18. Do's and Don't in Area of Operation

PC19. Safety Precaution Majors before Operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the Organization Chart, Relevant legislation, standards, policies, procedures & Code of Conduct followed in the company
- KU2.** Different Types of Products Manufactured by the Company
- KU3.** Quality Check and Grading and Segregation of non conforming Products
- KU4.** Risk and impact of not following defined procedures/work instructions
- KU5.** Types of documentation in organization and importance of the same
- KU6.** Records to be maintained and implications of non-maintenance of the same
- KU7.** Importance of housekeeping & good shop floor practices (eg. 3S & 5S)
- KU8.** Health, Safety and Environment guidelines
- KU9.** Impact of poor practices on health, safety and environment
- KU10.** Potential hazards and actions to minimize the same
- KU11.** Impact of various practices on cost, quality, productivity, delivery and safety Handover/ Takeover the equipment/ work area as per company SOP
- KU12.** The Emergency Stops Procedure for Extruder Machine
- KU13.** Cleanliness and safety requirements before and after Extruder operation
- KU14.** Importance of RPM, Temperature and Pressure Parameters
- KU15.** Process Parameters and its control
- KU16.** Common Faults and Trouble shooting
- KU17.** Health hazards of process and safety precautions
- KU18.** Measurement techniques using gauges and balance (for thickness, width and weight)
- KU19.** Response to emergencies e.g. Power failures, fire and system failures and manual intervention to avoid disaster
- KU20.** Knowledge of appropriate batch size with respect to appropriate machinery

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Fill up appropriate technical forms, process charts, activity logs in required format of the company

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- GS2.** Perform functional mathematical operations, including apply basic mathematical principles, such as numbers and space, and techniques such as estimation and approximation, for practical purposes
- GS3.** How to enter into the history card details of the fault identified in the plastic product manufactured
- GS4.** Read and understand manuals, Warning, health and safety instructions, Raw Material Labels, Components, memos, reports, job cards etc
- GS5.** Read images, graphs, diagrams
- GS6.** Understand the various coding systems as per company norms
- GS7.** Express Opinions or Information clearly
- GS8.** Discuss task lists, schedules, and work-loads with co-workers
- GS9.** Communicate with Supervisors Job & Issue related tasks
- GS10.** Communication between Upstream and Downstream teams
- GS11.** Avoid using jargon, slang or acronyms when communicating with a supervisor /fellow subordinates etc unless it is required
- GS12.** Work in a team and other behavioral skills required to support the small group activities (Quality Circle, Cross Functional Team, Suggestion Scheme)
- GS13.** Practice honesty with respect to company property and time
- GS14.** Take responsibility for completing one own work assignment
- GS15.** Take initiative to enhance/learn skills in one area of work
- GS16.** The capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse one learning.
- GS17.** Is open to new ways of doing things
- GS18.** The capacity to envisage and articulate personal goals to develop strategies and take action to achieve them.
- GS19.** Avoid absenteeism
- GS20.** Act objectively , rather than impulsively or emotionally when faced with difficult/stressful or emotional situations
- GS21.** Work in disciplined factory environment
- GS22.** Be punctual
- GS23.** Make decisions pertaining to the concerned area of work
- GS24.** Operating different types of extruders
- GS25.** Handling of Plastics Raw Materials, Additives and Master Batches
- GS26.** Handling of various types of material handling equipment like forklifts, trolley etc
- GS27.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS28.** Use common sense and make judgments during day to day basis.
- GS29.** Use reasoning skills to identify and resolve basic problems
- GS30.** Identify immediate or temporary solutions to resolve delays
- GS31.** DiagLOe common problems in the machine based on visual inspection, sound etc
- GS32.** Suggest improvements(if any) in process based on experience

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>HDPE / PVC Pipe Extruders and their parts</i>	16	27	-	-
PC1. Types of HDPE / PVC Extruders	3	3	-	-
PC2. Extruder Parts and Their Functions	3	4	-	-
PC3. Pressure and Vacuum Sizing Units	2	4	-	-
PC4. Types of Dies Used for different Extruded Pipes	3	4	-	-
PC5. Operations of Haul Off Units	1	4	-	-
PC6. Dismantling and assembling Extruder Parts	2	4	-	-
PC7. Safety Precaution taken during assembling and disassembling	2	4	-	-
<i>Extrusion Process and Parameters</i>	4	8	-	-
PC8. Common Process Parameter like Temperature, Pressure and Speed and its controls	2	4	-	-
PC9. Effect of process parameters on Product Properties	2	4	-	-
<i>Machine Operation and Controls</i>	8	15	-	-
PC10. Trial Production and checking product stabilization	2	4	-	-
PC11. Actual Production and Parameter / Process Control	2	4	-	-
PC12. Quality Check and Continuous Production	2	3	-	-
PC13. Post production and storing	2	4	-	-
<i>Common faults and remedies</i>	6	11	-	-
PC14. Common faults found and trouble shooting	2	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Segregation of faulty product and action taken	2	4	-	-
PC16. Disposal of faulty products as per laid down procedure	2	3	-	-
<i>Health & Safety</i>	6	9	-	-
PC17. Understanding Safety Equipments and Its Use	2	3	-	-
PC18. Do's and Don't in Area of Operation	2	3	-	-
PC19. Safety Precaution Majors before Operations	2	3	-	-
NOS Total	40	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0303
NOS Name	Perform HDPE /PVC Pipe Extruder Machine Operation
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Extrusion, Plastics Extrusion
NSQF Level	3.5
Credits	4
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0304: Perform - Plastic Film Extruder Machine Operation

Description

This NOS is about knowledge of Plastics Film Extrusion Process

Scope

The scope covers the following :

- This unit/task covers the following:
- Plastics Film Extruders and their parts
- Extrusion Process and Parameters.
- Machine Operation and Controls.
- Common faults and remedies.
- Health and Safety

Elements and Performance Criteria

Plastics Film Extruders and their parts

To be competent, the user/individual on the job must be able to:

- PC1.** Types of Film Extruders
- PC2.** Extruder Parts and Their Functions
- PC3.** Air Compressor Pressure Sizing Units
- PC4.** Types of Dies Used for different Extruded Films
- PC5.** Operations of Haul Off Units
- PC6.** Dismantling and assembling Extruder Parts
- PC7.** Safety Precaution taken during assembling and disassembling

Extrusion Process and Parameters

To be competent, the user/individual on the job must be able to:

- PC8.** Common Process Parameter like Temperature, Pressure and Speed and its controls
- PC9.** Effect of process parameters on Product Properties

Machine Operation and Controls

To be competent, the user/individual on the job must be able to:

- PC10.** Trial Production and checking product stabilization
- PC11.** Actual Production and Parameter / Process Control
- PC12.** Quality Check and Continuous Production.

Common faults and remedies

To be competent, the user/individual on the job must be able to:

- PC13.** Common faults found and trouble shooting
- PC14.** Segregation of faulty product and action taken
- PC15.** Disposal of faulty products as per laid down procedure

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Health & Safety

To be competent, the user/individual on the job must be able to:

PC16. Understanding Safety Equipments and Its Use

PC17. Do's and Don't in Area of Operation

PC18. Safety Precaution Majors before Operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the Organization Chart, Relevant legislation, standards, policies, procedures & Code of Conduct followed in the company
- KU2.** Different Types of Products Manufactured by the Company
- KU3.** Quality Check and Grading and Segregation of non conforming Products
- KU4.** Risk and impact of not following defined procedures/work instructions
- KU5.** Types of documentation in organization and importance of the same
- KU6.** Records to be maintained and implications of non-maintenance of the same
- KU7.** Importance of housekeeping & good shop floor practices (eg. 3S & 5S)
- KU8.** Health, Safety and Environment guidelines
- KU9.** Impact of poor practices on health, safety and environment
- KU10.** Potential hazards and actions to minimize the same
- KU11.** Impact of various practices on cost, quality, productivity, delivery and safety Handover/ Takeover the equipment/ work area as per company SOP
- KU12.** The Emergency Stops Procedure for Extruder Machine
- KU13.** Cleanliness and safety requirements before and after Extruder operation
- KU14.** Importance of RPM, Temperature and Pressure Parameters
- KU15.** Process Parameters and its control
- KU16.** Common Faults and Trouble shooting
- KU17.** Health hazards of process and safety precautions
- KU18.** Measurement techniques using gauges and balance (for thickness, width and weight)
- KU19.** Response to emergencies e.g. Power failures, fire and system failures and manual intervention to avoid disaster
- KU20.** Knowledge of appropriate batch size with respect to appropriate machinery

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Fill up appropriate technical forms, process charts, activity logs in required format of the company

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- GS2.** Perform functional mathematical operations, including apply basic mathematical principles, such as numbers and space, and techniques such as estimation and approximation, for practical purposes
- GS3.** How to enter into the history card details of the fault identified in the plastic product manufactured
- GS4.** Read and understand manuals, Warning, health and safety instructions, Raw Material Labels, Components, memos, reports, job cards etc
- GS5.** Read images, graphs, diagrams
- GS6.** Understand the various coding systems as per company norms
- GS7.** Express Opinions or Information clearly
- GS8.** Discuss task lists, schedules, and work-loads with co-workers
- GS9.** Communicate with Supervisors Job & Issue related tasks
- GS10.** Communication between Upstream and Downstream teams
- GS11.** Avoid using jargon, slang or acronyms when communicating with a supervisor /fellow subordinates etc unless it is required
- GS12.** Work in a team and other behavioral skills required to support the small group activities (Quality Circle, Cross Functional Team, Suggestion Scheme)
- GS13.** Practice honesty with respect to company property and time
- GS14.** Take responsibility for completing one own work assignment
- GS15.** Take initiative to enhance/learn skills in one area of work
- GS16.** The capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse one learning
- GS17.** Is open to new ways of doing things
- GS18.** The capacity to envisage and articulate personal goals to develop strategies and take action to achieve them
- GS19.** Avoid absenteeism
- GS20.** Act objectively , rather than impulsively or emotionally when faced with difficult/stressful or emotional situations
- GS21.** Work in disciplined factory environment
- GS22.** Be punctual
- GS23.** Make decisions pertaining to the concerned area of work
- GS24.** Operating different types of extruders
- GS25.** Handling of Plastics Raw Materials, Additives and Master Batches
- GS26.** Handling of various types of material handling equipment like forklifts, trolley etc
- GS27.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS28.** Use common sense and make judgments during day to day basis
- GS29.** Use reasoning skills to identify and resolve basic problems
- GS30.** Identify immediate or temporary solutions to resolve delays
- GS31.** DiagLOe common problems in the machine based on visual inspection, sound etc
- GS32.** Suggest improvements(if any) in process based on experience

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Plastics Film Extruders and their parts</i>	15	25	-	-
PC1. Types of Film Extruders	1	2	-	-
PC2. Extruder Parts and Their Functions	3	2	-	-
PC3. Air Compressor Pressure Sizing Units	3	4	-	-
PC4. Types of Dies Used for different Extruded Films	2	4	-	-
PC5. Operations of Haul Off Units	2	4	-	-
PC6. Dismantling and assembling Extruder Parts	2	4	-	-
PC7. Safety Precaution taken during assembling and disassembling	2	5	-	-
<i>Extrusion Process and Parameters</i>	6	10	-	-
PC8. Common Process Parameter like Temperature, Pressure and Speed and its controls	3	5	-	-
PC9. Effect of process parameters on Product Properties	3	5	-	-
<i>Machine Operation and Controls</i>	8	16	-	-
PC10. Trial Production and checking product stabilization	3	5	-	-
PC11. Actual Production and Parameter / Process Control	3	6	-	-
PC12. Quality Check and Continuous Production.	2	5	-	-
<i>Common faults and remedies</i>	6	12	-	-
PC13. Common faults found and trouble shooting	2	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Segregation of faulty product and action taken	2	4	-	-
PC15. Disposal of faulty products as per laid down procedure	2	4	-	-
<i>Health & Safety</i>	5	7	-	-
PC16. Understanding Safety Equipments and Its Use	2	4	-	-
PC17. Do's and Don't in Area of Operation	2	2	-	-
PC18. Safety Precaution Majors before Operations	1	1	-	-
NOS Total	40	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0304
NOS Name	Perform - Plastic Film Extruder Machine Operation
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Extrusion, Plastics Extrusion
NSQF Level	3.5
Credits	4
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0306: To - carry out reporting and documentation

Description

This unit is about carrying out reporting and documentation

Scope

The scope covers the following :

- This unit/task covers the following
- Reporting of data/problem/incidents etc
- Documentation
- Information Security

Elements and Performance Criteria

Reporting

To be competent, the user/individual on the job must be able to:

- PC1.** Report data/problems/incidents as per the laid down procedure in the prescribed format and registers
- PC2.** Report to the appropriate authority as laid down by the company procedure

Recording and Documentation

To be competent, the user/individual on the job must be able to:

- PC3.** Identify documentation to be completed relating to the job profile
- PC4.** Record details accurately in an appropriate format
- PC5.** Complete all documentation within stipulated time according to company procedure
- PC6.** Make sure documents are available to all appropriate authorities to inspect

Information Security

To be competent, the user/individual on the job must be able to:

- PC7.** Respond to requests for information in an appropriate manner whilst following organizational procedures
- PC8.** Inform the appropriate authority of requests for information received

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different methods of recording information
- KU2.** Various documents that need to be maintained
- KU3.** Company procedure for filling/maintaining up the documents
- KU4.** Procedures for reporting to the appropriate authority
- KU5.** Procedures for recording damage, breakages etc

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- KU6.** The importance of accurate documentation completion within a prescribed timeframe
- KU7.** The actions to be taken if the documents are not correct
- KU8.** The importance of maintaining the security and confidentiality of recorded information
- KU9.** Procedures to maintain confidentiality of information
- KU10.** The appropriate method for responding to requests for information
- KU11.** The reporting procedures to be followed before disclosing information to any outside party

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Construct simple sentences and express ideas clearly through written communication
- GS2.** Fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS3.** Write simple letters, mails, etc
- GS4.** Perform functional mathematical operations, including apply basic mathematical principles, such as numbers and space, and techniques such as estimation and approximation, for practical purposes
- GS5.** Read and understand manuals, health and safety instructions, memos, documents, reports, job cards etc
- GS6.** Read images, graphs, diagrams etc
- GS7.** Understand the various coding systems as per company norms
- GS8.**
 - Express statements, opinions or information clearly so that others can hear
 - and understand
- GS9.** Respond appropriately to any queries
- GS10.** Communicate with supervisor, upstream and downstream teams
- GS11.** Work in a team and other behavioral skills required to support the small group activities (Quality Circle, Cross Functional Team, Suggestion Scheme)
- GS12.** Practice honesty with respect to company property and time
- GS13.**
 - Communicate with people in a form and manner and using language that is
 - open and respectful
- GS14.** Resolve any difficulties in relationships with colleagues , or get help from an appropriate person, in a way that preserves goodwill and trust
- GS15.** Take responsibility for completing one own risk assignment
- GS16.** Take initiative to enhance/learn skills in one area of work
- GS17.** The capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse one learning.
- GS18.** Is open to new ways of doing things
- GS19.** The capacity to envisage and articulate personal goals to develop strategies and take action to achieve them
- GS20.** Avoid absenteeism

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- GS21.** Act objectively , rather than impulsively or emotionally when faced with difficult/stressful or emotional situations
- GS22.** Work in disciplined factory environment
- GS23.** Be punctual
- GS24.** Operating different types of extruders
- GS25.** Handling of rubber compound & chemicals
- GS26.** • Handling of various types of material handling equipment like forklifts, trolley
• capacity to apply technology, combining the physical and sensory skills needed to
• operate equipment with the understanding of scientific and technological
• principles needed to explore and adapt systems
- GS27.** seek clarification on problems from others
- GS28.** apply problem-solving approaches in different situations
- GS29.** refer anomalies to the line manager
- GS30.** Interpret quality for sheet
- GS31.** Suggest improvements(if any) in process/product/materials based on results and experience
- GS32.** Proper collection of waste material
- GS33.** Identify defects in the material and communicate it at the earliest and suggest improvements(if any) in process/material based on experience
- GS34.** Diagnose common problems in the machine based on visual inspection, sound , temperature etc
- GS35.** Suggest improvements(if any) in process based on experience
- GS36.** Handle equipment/rubber sheet
- GS37.** apply problem-solving approaches in different situations
- GS38.** refer anomalies to the line manager

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Reporting</i>	2	6	-	-
PC1. Report data/problems/incidents as per the laid down procedure in the prescribed format and registers	1	2	-	-
PC2. Report to the appropriate authority as laid down by the company procedure	1	4	-	-
<i>Recording and Documentation</i>	6	16	-	-
PC3. Identify documentation to be completed relating to the job profile	1	4	-	-
PC4. Record details accurately in an appropriate format	2	4	-	-
PC5. Complete all documentation within stipulated time according to company procedure	1	4	-	-
PC6. Make sure documents are available to all appropriate authorities to inspect	2	4	-	-
<i>Information Security</i>	2	8	-	-
PC7. Respond to requests for information in an appropriate manner whilst following organizational procedures	1	4	-	-
PC8. Inform the appropriate authority of requests for information received	1	4	-	-
NOS Total	10	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0306
NOS Name	To - carry out reporting and documentation
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Extrusion, Plastics Extrusion
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0307: To carryout basic quality check of finished products with reference to approved product

Description

This NOS is about carrying out quality control activities

Scope

The scope covers the following :

- This unit/task covers the following:
- Inspection: Carrying out quality checks to identify problems
- Analysis: Take corrective actions
- Reporting the results

Elements and Performance Criteria

Inspection

To be competent, the user/individual on the job must be able to:

- PC1.** Ensure that total range of checks as per the prescribed national and International standards on regular intervals throughout the shifts.
- PC2.** Use appropriate measuring instruments, equipment, tools, accessories etc, as prescribed / required

Analysis

To be competent, the user/individual on the job must be able to:

- PC3.** Identify non-conformities to quality assurance standards
- PC4.** Identify potential causes of non-conformities to quality assurance standards
- PC5.**
 - Identify impact on final product due to non-conformance to prescribed
 - Standards
- PC6.** Evaluating the need for action to ensure that problems do not reoccur
- PC7.** Suggest corrective action to address problem
- PC8.** Review effectiveness of corrective action

Reporting

To be competent, the user/individual on the job must be able to:

- PC9.** Interpret the results of the quality check correctly
- PC10.** Take up results of the findings with QC in charge/appropriate authority
- PC11.** Take up the results of the findings within stipulated time
- PC12.** Record of results of action taken
- PC13.** Record adjustments not covered by established procedures for future reference
- PC14.** Review effectiveness of action taken
- PC15.** Follow reporting procedures where the cause of defect cannot be identified

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The importance of quality control procedures with respect to national and international standards
- KU2.** Relevance and importance of activities and how they contribute to the achievement of the quality objectives
- KU3.** Proper procedure for selecting the material/product and performing quality checks without affecting the material
- KU4.** Availability of work instructions, as necessary
- KU5.** Characteristics of the product/material
- KU6.** Use of suitable equipment
- KU7.** Availability and use of monitoring and measuring devices
- KU8.** Requirements of records
- KU9.** Importance of maintaining accurate up-to-date records
- KU10.** The need to report within the stipulated time
- KU11.** Implications of inaccurate measuring and testing instruments and equipment
- KU12.** The cost of non-conformance to quality standards
- KU13.**
 - Implications (impact on internal/external customers) of defective products,
 - materials or components

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
 - Construct simple sentences and express ideas clearly through written
 - communication
- GS2.**
 - Fill up appropriate technical forms, process charts, activity logs in required
 - format of the company
- GS3.** Write simple letters, mails, etc
- GS4.** Perform functional mathematical operations, including apply basic mathematical principles, such as numbers and space, and techniques such as estimation and approximation, for practical purposes
- GS5.** Read and understand manuals, health and safety instructions, memos, reports, job cards etc
- GS6.** Read images, graphs, diagrams
- GS7.** Understand the various coding systems as per company norms
- GS8.**
 - Express statements, opinions or information clearly so that others can hear
 - and understand
- GS9.** Respond appropriately to any queries
- GS10.** Communicate with supervisor
- GS11.** Communicate with upstream and downstream teams

Qualification Pack

- GS12.** Work in a team and other behavioral skills required to support the small group activities (Quality Circle, Cross Functional Team, Suggestion Scheme)
- GS13.** Practice honesty with respect to company property and time
- GS14.** • Communicate with people in a form and manner and using language that is
• open and respectful
- GS15.** Resolve any difficulties in relationships with colleagues , or get help from an appropriate person, in a way that preserves goodwill and trust
- GS16.** Take responsibility for completing one own work assignment
- GS17.** Take initiative to enhance/learn skills in one area of work
- GS18.** The capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse one learning
- GS19.** Is open to new ways of doing things
- GS20.** The capacity to envisage and articulate personal goals to develop strategies and take action to achieve them
- GS21.** Avoid absenteeism
- GS22.** • Act objectively , rather than impulsively or emotionally when faced with
• difficult/stressful or emotional situations
- GS23.** Work in disciplined factory environment
- GS24.** Be punctual
- GS25.** Operating different types of measuring instruments
- GS26.** Operating of different types of Testing equipments
- GS27.** Handling of various types of material handling equipment like forklifts, trolley
- GS28.** The capacity to apply technology, combining the physical and sensory skills needed to operate equipment with the understanding of scientific and technological principles needed to explore and adapt systems
- GS29.** Seek clarification on problems from others
- GS30.** Apply problem-solving approaches in different situations
- GS31.** Refer anomalies to the line manager
- GS32.** Interpret quality for sheet
- GS33.** Suggest improvements(if any) in process/product/materials based on results and experience
- GS34.** Proper collection of waste material
- GS35.** Identify defects in the material and communicate it at the earliest and suggest improvements (if any) in process/material based on experience
- GS36.** DiagLOe common problems in the machine based on visual inspection, sound , temperature etc
- GS37.** Suggest improvements(if any) in process based on experience

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspection</i>	2	8	-	-
PC1. Ensure that total range of checks as per the prescribed national and International standards on regular intervals throughout the shifts.	1	4	-	-
PC2. Use appropriate measuring instruments, equipment, tools, accessories etc, as prescribed / required	1	4	-	-
<i>Analysis</i>	12	32	-	-
PC3. Identify non-conformities to quality assurance standards	1	4	-	-
PC4. Identify potential causes of non-conformities to quality assurance standards	2	4	-	-
PC5. • Identify impact on final product due to non-conformance to prescribed • Standards	2	6	-	-
PC6. Evaluating the need for action to ensure that problems do not reoccur	3	6	-	-
PC7. Suggest corrective action to address problem	2	6	-	-
PC8. Review effectiveness of corrective action	2	6	-	-
<i>Reporting</i>	16	40	-	-
PC9. Interpret the results of the quality check correctly	2	6	-	-
PC10. Take up results of the findings with QC in charge/appropriate authority	2	6	-	-
PC11. Take up the results of the findings within stipulated time	2	6	-	-
PC12. Record of results of action taken	2	6	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Record adjustments not covered by established procedures for future reference	3	6	-	-
PC14. Review effectiveness of action taken	2	6	-	-
PC15. Follow reporting procedures where the cause of defect cannot be identified	3	4	-	-
NOS Total	30	80	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0307
NOS Name	To carryout basic quality check of finished products with reference to approved product
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Extrusion, Plastics Extrusion
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S

Description

This unit is about creating a Safe and Healthy work place, adhering to the safety guidelines in the working area, following practices which are not impacting the environment in a negative manner. This NOS is about ensuring all 5S activities both at the shop floor and the office area to facilitate increase in work productivity. It covers knowledge of fire safety, common first aid applications, safe practice and emergency applications.

Scope

The scope covers the following :

- The role holder will be responsible for
- Identifying and reporting of risks
- Creating and sustaining a safe, clean and environment friendly Work place.
- Ensure sorting, streamlining & organizing, storage and documentation, cleaning, standardization and sustenance across the plant and office premises of the organization

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** Wear protective clothing/equipment for specific tasks and work conditions
- PC2.** Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.
- PC3.** Apply good housekeeping practices at all times

Fire safety

To be competent, the user/individual on the job must be able to:

- PC4.** Use the various appropriate fire extinguishers on different types of fires correctly
- PC5.** Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.

Identify and report the risks identified

To be competent, the user/individual on the job must be able to:

- PC6.** Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.
- PC7.** Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.
- PC8.** Create awareness amongst other by sharing information on the identified risks.

Ensure sorting

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To be competent, the user/individual on the job must be able to:

- PC9.** Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.
- PC10.** Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions
- PC11.** Follow the technique of waste disposal and waste storage in the proper bins as per SOP
- PC12.** Segregate the items which are labeled as red tag items for the process area and keep them in the correct places
- PC13.** Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions
- PC14.** Ensure that areas of material storage areas are not overflowing
- PC15.** Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required
- PC16.** Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area
- PC17.** Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards

Ensure proper documentation and storage(organizing, streamlining)

To be competent, the user/individual on the job must be able to:

- PC18.** Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists
- PC19.** Check that the items in the respective areas have been identified as broken or damaged
- PC20.** Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.
- PC21.** Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Relevant standards, procedures and policies related to Health, Safety and Environment followed in the company
- KU2.** Emergency handling procedures & hierarchy for escalation
- KU3.** Basic knowledge of Safety procedures(firefighting, first aid) within the organization
- KU4.** Basic knowledge of various types of PPEs and their usage
- KU5.** Basic knowledge of risks/hazards associated with each occupation in the organization
- KU6.** Knowledge of personal hygiene and how an individual can contribute towards creating a highly safe and clean working environment
- KU7.** Meaning of hazards and risks
- KU8.** Health and safety hazards commonly present in the work environment and related precautions

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- KU9.** Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU10.** Possible causes of risk and accident (due to oil leakage)
- KU11.** Safe working practices when working with tools and machines
- KU12.** Safe working practices while working at various hazardous sites
- KU13.** Where to find all the general health and safety equipment in the workplace
- KU14.** Various dangers associated with the use of electrical equipment
- KU15.** Preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU16.** Importance of using protective clothing/equipment while working
- KU17.** Precautionary activities to prevent the fire accident
- KU18.** Various causes of fire
- KU19.** Techniques of using the different fire extinguishers
- KU20.** Different methods of extinguishing fire
- KU21.** Different materials used for extinguishing fire
- KU22.** Rescue techniques applied during a fire hazard
- KU23.** Various types of safety signs and what they mean
- KU24.** Appropriate basic first aid treatment relevant to the condition eg. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU25.** Content of written accident report
- KU26.** Potential injuries and ill health associated with incorrect manual handling
- KU27.** Safe lifting and carrying practices
- KU28.** Personal safety, health and dignity issues relating to the movement of a person by others
- KU29.** Potential impact to a person who is moved incorrectly
- KU30.** Have basic knowledge of 5S procedures
- KU31.** Know various types 5s practices followed in various areas
- KU32.** Understand the 5S checklists provided in the department/ team
- KU33.** Have skills to identify useful & non useful items
- KU34.** Have knowledge of labels , signs & colors used as indicators
- KU35.** Have knowledge on how to sort and store various types of tools, equipment, material etc.
- KU36.** Know , how to identify various types of waste products
- KU37.** Understand the impact of waste/ dirt/ dust/unwanted substances on the process/ environment/ machinery/ human body.
- KU38.** Have knowledge of best ways of cleaning & waste disposal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write basic level notes and observations
- GS2.** Read safety instructions put up across the plant premises

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- GS3.** Read safety precautions mentioned in equipment manuals and panels to understand the potential risks associated
- GS4.** Effectively communicate information to team members
- GS5.** Inform employees in the plant and concerned functions about events, incidents & potential risks observed related to Safety, Health and Environment.
- GS6.** Question operator/ supervisor in order to understand the safety related issues
- GS7.** Attentively listen with full attention and comprehend the information given by the speaker during safety drills and training programs
- GS8.** Plan and organize the work order and jobs received from the internal customers
- GS9.** Plan and organize the design documents received from internal customers
- GS10.** Organize all process/ equipment manuals so that sorting out information is Fast
- GS11.** Use common sense and make judgments during day to day basis
- GS12.** Use reasoning skills to identify and resolve basic problems
- GS13.** Use intuition to detect any potential problems which could arise during operations
- GS14.** Follow instructions and work on areas of improvement identified
- GS15.** Complete the assigned tasks with minimum supervision
- GS16.** Complete the job defined by the supervisor within the timelines and quality norms

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	1.5	6	-	-
PC1. Wear protective clothing/equipment for specific tasks and work conditions	0.5	2	-	-
PC2. Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.	0.5	2	-	-
PC3. Apply good housekeeping practices at all times	0.5	2	-	-
<i>Fire safety</i>	1	4	-	-
PC4. Use the various appropriate fire extinguishers on different types of fires correctly	0.5	2	-	-
PC5. Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.	0.5	2	-	-
<i>Identify and report the risks identified</i>	1.5	6	-	-
PC6. Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.	0.5	2	-	-
PC7. Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.	0.5	2	-	-
PC8. Create awareness amongst other by sharing information on the identified risks.	0.5	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure sorting</i>	4	10	-	-
PC9. Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.	0.5	2	-	-
PC10. Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions	0.5	2	-	-
PC11. Follow the technique of waste disposal and waste storage in the proper bins as per SOP	0.5	1	-	-
PC12. Segregate the items which are labeled as red tag items for the process area and keep them in the correct places	0.5	1	-	-
PC13. Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions	0.5	1	-	-
PC14. Ensure that areas of material storage areas are not overflowing	0.25	0.5	-	-
PC15. Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required	0.25	0.5	-	-
PC16. Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area	0.5	1	-	-
PC17. Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards	0.5	1	-	-
<i>Ensure proper documentation and storage(organizing, streamlining)</i>	2	4	-	-
PC18. Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists	0.5	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. Check that the items in the respective areas have been identified as broken or damaged	0.5	1	-	-
PC20. Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.	0.5	1	-	-
PC21. Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions	0.5	1	-	-
NOS Total	10	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0401
NOS Name	Basic health and safety practices at the workplace, 5S
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

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PC6. work with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

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- KU8.** different types of disabilities and appropriate communication and behaviour towards PwD
- KU9.** different types of financial products and services
- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

CPC/CAP/N0305: To Carryout House Keeping

Description

This NOS is about providing a basic knowledge of House Keeping In the Raw Material and Finished Items Storage and Operational Area

Scope

The scope covers the following :

- This unit/task covers the following:
- Preparing for housekeeping activities
- Carry out housekeeping activities
- Post housekeeping activities

Elements and Performance Criteria

Pre housekeeping activities

To be competent, the user/individual on the job must be able to:

- PC1.** Take an overlook of the Area under House Keeping
- PC2.** Put appropriate Signage immediately if oily substance / Water spills on the floor to avoid accident
- PC3.** If certain housekeeping activities require to be performed by housekeeping staffs, the Inform them
- PC4.** If it has to be carried out by self then, Identify the material / equipment required for cleaning the areas
- PC5.** Plan the sequence for cleaning the area to avoid re-soiling the cleaned areas and surfaces
- PC6.** Display the appropriate signage for the work being conducted
- PC7.** Ensure that there is adequate ventilation for the work being carried out
- PC8.** Wear the personal protective equipment required for the cleaning method and materials being used

Operations

To be competent, the user/individual on the job must be able to:

- PC9.** With right cleaning process carry out cleaning activities without disturbing others
- PC10.** Report to the appropriate person if any difficulties in carrying out your work

Post housekeeping activities

To be competent, the user/individual on the job must be able to:

- PC11.** Ensure that there is no oily substance / Water spill on the floor, If found the put the Signage immediately to avoid accident
- PC12.** Follow workplace procedures to deal with any accidental damage caused during the cleaning process
- PC13.** Ensure that, on completion of the work, the area is left clean and dry and free from any leftover or scrap

Qualification Pack

PC14. Return the equipment, materials and personal protective equipment that were used to the right places and check the inventory for the next cycle

PC15. Dispose the waste garnered from the activity in an appropriate manner

General

To be competent, the user/individual on the job must be able to:

PC16. Maintain schedules and records for housekeeping duty

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Adequate knowledge about the levels of hygiene required at the workplace and why it is important to maintain them during your work

KU2. Type of Cleaning Materials / Equipments and its uses for different cleaning needs

KU3.

- The right method of using cleaning equipment / machinery used during
- Housekeeping

KU4. The importance of appropriate personal protection like gloves, goggles and masks etc for handling different materials and chemicals

KU5. The correct sequence for cleaning the work area

KU6. The importance & time taken by the treatment to work if any

KU7. Process of cleaning the surfaces without causing injury or damage

KU8.

- The method to check the treated surface and equipment on completion of
- Cleaning

KU9. Procedures for reporting any unidentified soiling.

KU10. Procedures for disposing off waste

KU11. Procedures for disposing off or storing personal protective equipment

KU12. Escalation procedures for soils or stains that could not be removed

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1.

- Construct simple sentences and express ideas clearly through written
- communication

GS2. Fill up appropriate technical forms, activity logs in required format of the company

GS3. Write simple letters, mails, etc

GS4. Read and understand manuals, health and safety instructions, memos, reports, job cards etc

GS5. Read Signage, Images, graphs, diagrams

GS6. Understand the various coding systems as per company norms

GS7. Express statements, opinions or information clearly so that others can hear and understand

GS8. Respond appropriately to any queries

GS9. Communicate with supervisor, colleagues and downstream people

Qualification Pack

- GS10.** Work in a team and other behavioral skills required to support the small group activities (Quality Circle, Cross Functional Team, Suggestion Scheme)
- GS11.** Practice honesty with respect to company property and time Communicate with people in a form and manner and using language that is open and respectful
- GS12.** Resolve any difficulties in relationships with colleagues , or get help from an appropriate person, in a way that preserves goodwill and trust
- GS13.** Take responsibility for completing one work assignment
- GS14.** Take initiative to enhance/learn skills in one area of work
- GS15.** Avoid absenteeism and be Punctual
- GS16.** Act objectively , rather than impulsively or emotionally when faced with difficult/stressful or emotional situations
- GS17.** Work in disciplined factory environment
- GS18.** Operating different types Cleaning Machines and Equipments
- GS19.** Handling of different cleaning materials and chemicals
- GS20.** Taking Immediate decision for unwanted mess-up of scrap or other issues related to housekeeping
- GS21.** Observe, Inspect before taking up the job.
- GS22.** Seek clarification or assistance on problems if any from others
- GS23.** Apply problem-solving approaches in different situations
- GS24.** Suggest improvements (if any) in process/product/materials based on results and experience
- GS25.** Proper segregation and collection of waste material
- GS26.** Identify the issues for generation of waste if more than the prescribed limit and communicate it at the earliest and suggest improvements(if any) in process/material based on experience

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre housekeeping activities</i>	5	9	-	-
PC1. Take an overlook of the Area under House Keeping	0.5	1	-	-
PC2. Put appropriate Signage immediately if oily substance / Water spills on the floor to avoid accident	0.5	1	-	-
PC3. If certain housekeeping activities require to be performed by housekeeping staffs, the Inform them	0.5	1	-	-
PC4. If it has to be carried out by self then, Identify the material / equipment required for cleaning the areas	0.5	1	-	-
PC5. Plan the sequence for cleaning the area to avoid re-soiling the cleaned areas and surfaces	0.5	1	-	-
PC6. Display the appropriate signage for the work being conducted	0.5	1	-	-
PC7. Ensure that there is adequate ventilation for the work being carried out	1	1	-	-
PC8. Wear the personal protective equipment required for the cleaning method and materials being used	1	2	-	-
<i>Operations</i>	2	4	-	-
PC9. With right cleaning process carry out cleaning activities without disturbing others	1	2	-	-
PC10. Report to the appropriate person if any difficulties in carrying out your work	1	2	-	-
<i>Post housekeeping activities</i>	2.5	6	-	-
PC11. Ensure that there is no oily substance / Water spill on the floor, If found the put the Signage immediately to avoid accident	0.5	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Follow workplace procedures to deal with any accidental damage caused during the cleaning process	0.5	1	-	-
PC13. Ensure that, on completion of the work, the area is left clean and dry and free from any leftover or scrap	0.5	1	-	-
PC14. Return the equipment, materials and personal protective equipment that were used to the right places and check the inventory for the next cycle	0.5	1	-	-
PC15. Dispose the waste garnered from the activity in an appropriate manner	0.5	1	-	-
<i>General</i>	0.5	1	-	-
PC16. Maintain schedules and records for housekeeping duty	0.5	1	-	-
NOS Total	10	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0305
NOS Name	To Carryout House Keeping
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Plastics Extrusion, Plastics Extrusion
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches are assigned to Training Assessment Wing (TAW), CIPET HO for planning of assessment
- Training Centers request TAW for Assessment and Certification of Trainees
- TAW identifies suitable assessor and nominates the assessor to the respective Training Centre
- TAW monitors the assessment process
- Training Centers maintain necessary records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 02 Assessors in a day (or) 01 assessor in 2 days

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- Check that the allotted time to the candidates to complete the Theory & Practical Assessment

3. Assessment Quality Assurance levels/Framework:

- Question bank / Question Paper is prepared by the Subject Matter Experts (SME) / Assessor
- Questions are mapped to the specified assessment criteria
- Certified Assessor & Trainer will be engaged in the process

4. Types of evidence or evidence-gathering protocol:

- Date / Time recorded for the reporting of the assessor from assessment location
- Assessment batch - Group Photo of Trainees along with Assessor

5. Method of verification or validation:

- Surprise visit to the assessment location
- Virtual meet with the Assessor / Trainees

6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored, soft copies of assessment evidences are stored in Email for future correspondence

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CPC/CAP/N0301.Understand the Basics of Plastics Raw Material & concept of Extrusion Techniques	20	40	-	-	60	10
CPC/CAP/N0302.Understand the basics & importance of Plastics Compounding /Mixing	30	50	-	-	80	10

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CPC/CAP/N0303.Perform HDPE /PVC Pipe Extruder Machine Operation	40	70	-	-	110	15
CPC/CAP/N0304.Perform - Plastic Film Extruder Machine Operation	40	70	-	-	110	15
CPC/CAP/N0306.To - carry out reporting and documentation	10	30	-	-	40	10
CPC/CAP/N0307.To carryout basic quality check of finished products with reference to approved product	30	80	-	-	110	15
CPC/CAP/N0401.Basic health and safety practices at the workplace, 5S	10	30	-	-	40	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
CPC/CAP/N0305.To Carryout House Keeping	10	20	-	-	30	5
Total	210	420	-	-	630	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.